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MILITARY SPECIFICATION

MODULE, FIBER OPTIC, RECEIVER, DIGITAL, 160 MBD, ASSOCIATED DETAIL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24791.

1. SCOPE

1.1 Scope. This specification, in conjunction with MIL-M-24791, describes a digital fiber optic receiver for use in SAFENET LAN installations. It may also be used in other applications where the modulation rate does not exceed 160 MBd (megabaud).

1.2 Classification. This specification applies to two types of receivers.

Type LS - for use in SAFENET class one and class two systems
Type HS - for use in SAFENET class three systems

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-49291/6 - Fiber Optical, 62.5/125 Micrometers, Radiation Harden (Metric), Associated Detail Specification For

MIL-H-38534 - Hybrid Microcircuits, General Specification For

MIL-M-24791 - Module, Fiber Optic, Transmitter or Receiver, Digital, General Requirements For

STANDARDS

MILITARY

MIL-STD-883 - Test Methods And Procedures For Microelectronics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command, SEA 05Q42, Washington, DC 20362-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-H-24793 with reference fiber, power level, and transition duration as specified in subparagraphs to this paragraph.

3.2.3.1 Reference fiber. The reference fiber is that described by MIL-F-49291/6.

3.2.3.2 Power level. Power level for:

Type LS receiver: -31 dBm to -14 dBm
Type HS receiver: -35 dBm to -10 dBm

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3.2.3.3 Transition duration. Transition durations to 0.65 UI for modulation rates between 125 MBd and 160 MBd.

3.2.4 Signal transfer.

3.2.4.1 Envelope waveforms to data output(s). The relationship between the input signal envelope waveform and each of the data output signals shall be as specified in the subparagraphs to this paragraph.

3.2.4.1.1 Phasing. The data outputs shall be complementary. Table I indicates the phase of the signal at the DATA outputs with respect to that of the envelope waveform at the input.

TABLE I. Phasing.

Level 1/ 2/		
Envelope waveform	Output logic	
	+DATA	-DATA
T	H	L
B	L	H

1/ "H" AND "L" indicate HIGH and LOW logic level, respectively (see 3.2.2).
2/ "T" and "B" indicate top and base level, respectively (see 3.2.3).

3.2.4.1.2 Bit error ratio. The bit error ratio (BER) of each of the data output signals relative to the envelope waveform at the input signal shall be specified in the subparagraphs to this paragraph. Averaging time for power measurement shall be 1 ms or greater.

3.2.4.1.2.1 Sampling instant. BER values apply for any sample location within ±15 percent of the center of the undistorted unit interval.

3.2.4.1.2.2 Bit pattern. BER values apply for the bit pattern shown in figure 1.

3.2.4.1.2.3 Sensitivity. BER shall not be greater than 2.5×10^{-10} for the following input power (P_i) levels:

For type LS: $-31 \text{ dBm} \leq P_i < -29 \text{ dBm}$
For type HS: $-35 \text{ dBm} \leq P_i < -33 \text{ dBm}$

3.2.4.1.2.4 Improvement. BER shall not be greater than 1.0×10^{-12} for the following input power (P_i) levels:

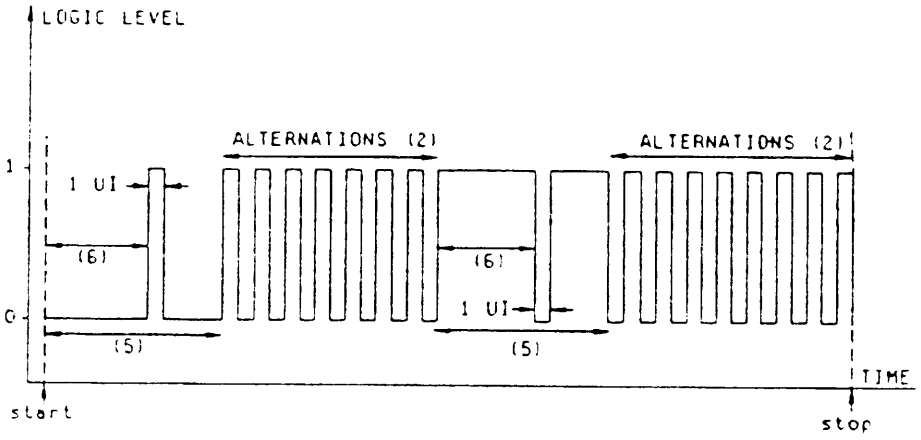
For type LS: $-29 \text{ dBm} \leq P_i \leq -14 \text{ dBm}$
For type HS: $-33 \text{ dBm} \leq P_i \leq -10 \text{ dBm}$

3.2.4.1.3 Optoelectronic propagation delay. Optoelectronic propagation delay between transitions of the envelope waveform at the optical interface and corresponding transitions at the data outputs shall be as specified in the subparagraphs to this paragraph.

3.2.4.1.3.1 Magnitude. Optoelectronic propagation delay associated with any input transition polarity (see 6.3) shall be not greater than 50 ns.

3.2.4.1.3.2 Difference. The difference between the optoelectronic delays associated with each input transition polarity shall be not greater than 0.4 ns when the envelope waveform pattern is a square wave.

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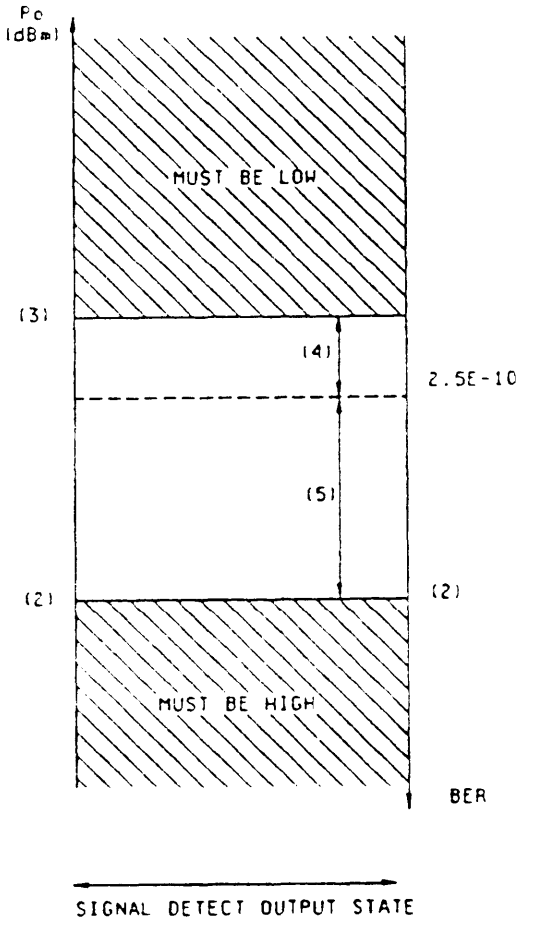


NOTES:
1. Modulation rate is maximum specified.
2. During alternations, the logic levels between "1" and "0" in each succeeding unit interval. The duty factor for the two alternation sequences, combined, is 0.5. The combined length of both alternation sequences equals one half the length of the pattern.
3. The pattern is repeated continuously, with the stop time of one pattern being the start time of the succeeding pattern.
4. UI= Unit interval.
5. Maximum required run length.
6. One half required maximum run length.

Figure 1. BER test pattern.

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3.2.4.2 Optical input to signal detect output. The relationship between the input signal and the signal detect output signal shall be as specified in the subparagraphs to this paragraph. These requirements are summarized in figure 2.



NOTES:
1. Po is the average power at the optical input to the receiver.
2. The greater of Po = -49 dBm or the input power producing a BER of 5E-6.
3. -35 dBm for type HS; -31 dBm for type LS.
4. Range within which signal detect output may change from HIGH to LOW; see note 6.
5. Range within which signal detect output may change from LOW to HIGH; see note 6.
6. The difference between the power levels at which signal detect changes state shall be greater than 0.75 dB.

Figure 2. Signal detect response.

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3.2.4.2.1 Input parameters detected. Signal detect shall respond to an input power level averaged over an interval greater than 400 ns.

3.2.4.2.2 Response.

3.2.4.2.2.1 Acceptable signal. Output shall be at a logic HIGH level when input power is greater than or equal to -31 dBm for type LS devices, and -35 dBm for type HS devices.

3.2.4.2.2.2 Unacceptable signal. Output shall be at a logic LOW when input power is less than that associated with BER values greater than or equal to 10^{-6} (see 3.2.4.1.2), or -49 dBm, whichever is greater.

3.2.4.2.2.3 Hysteresis. The difference between the input powers at which signal detect changes state from LOW to HIGH and from HIGH to LOW shall not be less than 0.75 dB.

3.2.4.2.2.4 Time. Signal detect shall provide the output level specified in 3.2.4.2.2.1 and 3.2.4.2.2.2 within 50 µs of the input achieving the associated condition. There shall be no response to change in input condition within the 100 µs interval following a change in output level.

3.2.5 Jitter.

3.2.5.1 Applicable conditions.

3.2.5.1.1 Reference waveform. Jitter on either DATA output shall be determined with respect to corresponding envelope waveform input transitions.

3.2.5.1.2 Reference magnitudes. The reference magnitude for the input envelope waveform shall be its associated mesial level. The reference magnitude for either DATA output shall be that associated with the designated logic interface (see 3.2.2).

3.2.5.1.3 Frequency range. Unless otherwise specified, all jitter spectral components within the range extending from 10 Hz to a frequency equal to one half the modulation rate of the envelope waveform are to be included.

3.2.5.1.4 Transition durations. Requirements apply for all allowed transition durations.

3.2.5.2 Random (see 6.3.3) and systematic (see 6.3.4). The subparagraphs to this paragraph apply for a square wave pattern at any allowed modulation rate and to all random and systematic components.

3.2.5.2.1 Rms. The increase in jitter shall be no greater than 170 ps rms.

3.2.5.2.2 Spectral distribution. Jitter spectral density from 10 Hz to 1.3 MHz shall be not greater than 4.3×10^4 fs^2/Hz.

3.2.5.3 Pattern dependent (see 6.3.2). The increase in pattern dependent jitter shall be no greater than 0.5 ns peak for an input signal pattern which is identified in annex A of ISO/IEC standard 9314-3.

3.2.5.4 Transfer to data output (see 6.3.2). The subparagraphs to this paragraph specify the relationship between certain specified parameters and jitter at both of the data outputs.

3.2.5.4.1 From envelope waveform. The magnitude of the transfer coefficient (see 6.3.1) relating jitter of output transitions to corresponding transitions of the input envelope waveform shall be no greater than 1.1.

3.2.5.4.2 From power supply inputs. The magnitude of the transfer coefficient (see 6.3.1) relating fluctuations of the voltage at any power supply terminal to the resulting jitter of signal at either data outputs shall be not greater than 1.5 ns/v for sinusoidal variations to 50 mv peak.

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3.2.6 Power supply. The receiver shall operate from a single voltage power supply.

3.2.6.1 Voltage. The functional requirements herein shall apply for power supply voltages from 4.75 to 5.25 V dc, inclusive, at V_{CC} with respect to V_{EE}.

3.2.6.2 Current. The current drawn from the power supply shall not be greater than 250 mA dc exclusive of the current supplied to output loads.

3.2.6.3 Ripple and noise. The functional requirements shall apply for sinusoidal variations of the power supply voltage to 50 mV peak.

3.2.7 Short circuit protection. Short circuit protection is not required for any terminal.

3.3 Environmental. Environmental requirements of MIL-M-24791 shall apply as specified and as supplemented by the subparagraph to this paragraph.

3.3.1 Operating temperature range. Case operating temperature range 3 of MIL-M-24791 shall apply.

3.4 Design and construction. Design and construction requirements of MIL-M-24791 shall apply as specified therein and modified or supplemented by the subparagraphs to this paragraph.

3.4.1 Package style. Package style AA shall be used as specified in MIL-M-24791, Appendix C.

3.4.2 Package isolation. There shall be no dc electrical connection between the package and any power supply terminal. Functional requirements herein shall be independent of connection of either of these terminals to the package.

3.4.3 Electrical terminals. Electrical connections shall be made through electrical terminals specified in this paragraph and its subparagraphs.

3.4.3.1 Assignment. Electrical terminal assignment shall be as specified in table II.

TABLE II. Terminal assignment.

Terminal no.	Description	Terminal no.	Description
1	V_{EE}	9	V_{EE}
2	- Data	10	No pin
3	+ Data	11	Case
4	V_{CC}	12	V_{CC}
5	V_{CC}	13	V_{EE}
6	Not connected 1/	14	+ Sig detect
7	V_{EE}	15	Unassigned 1/
8	V_{EE}	16	V_{EE}

NOTE: 1/ Unassigned or not connected pins may be used for other functions.

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3.4.3.2 Type. The electrical terminal type shall be as specified for the package style in MIL-M-24791, Appendix C (see 3.4.1).

3.4.3.3 Finish. Electrical terminal finish shall be gold plate in accordance with MIL-M-38534.

3.5 Reliability. The mean time between failure (MTBF), as calculated using MIL-HDBK-217, shall be greater than 10^5 hours at 25 °C.

3.6 Maintainability. The requirement stated in MIL-M-24791 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. All quality assurance provisions of MIL-M-24791 apply as either modified or supplemented as specified herein.

4.2 Quality assurance program. Quality assurance provisions of MIL-M-24791 apply.

4.3 Inspection and controls during manufacture. Inspection and controls during manufacture provisions of MIL-M-24791 apply.

4.4 Device screening. Device screening of MIL-M-24791 shall apply except that the stabilization bake is not required.

4.5 Quality conformance inspections (QCI). The provisions of MIL-M-24791 are supplemented and modified as specified.

4.5.1 Group A inspection. The group A inspections specified in table III shall be performed at the minimum, +5.0 V dc, and the maximum power supply voltages at the minimum, at +25°C, and at the maximum case operating temperature.

TABLE III. Group A inspections.

Parameter	Requirement paragraph	Test paragraph
Jitter/BER	3.2.4.1.2.3 / 3.2.5.2.1	4.7.7.5
Optoelectronic propagation delay difference	3.2.4.1.3.2	4.7.7.4
Signal detect	3.2.4.2	4.7.7.6
Power supply current	3.2.6.2	4.7.7.12

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4.5.1.1 Sample sizes and accept/reject criteria. Table IV specifies samples sizes and the number of allowed failures per inspection lot for QCI for a lot size of 50. Acquisition documentation must reference this table or provide an alternative.

Table IV. Group A sample sizes and allowed failures per inspection lot.

Parameter	Sample size	Allowed failures per lot
Jitter/BER	50	see notes 1/, 3/
Optoelectronic propagation delay difference	10	0; see note 2/
Signal detect	50	see note 1/
Power supply current	50	see note 1/

NOTES:
1/ Group A QCI Inspection Failure of MIL-M-24791 applies.
2/ Group A QCI Inspection Failure of MIL-M-24791 applies except that failure of any device does affect the remainder of the inspection lot as with a Group B, C or D inspection. QCI resubmission of MIL-M-24791 applies.
3/ Five devices shall be used to establish direct measurement correlation (see 4.7.5.5.1) and pattern sensitivity (see 4.7.5.5.2).

4.5.2 Group B, C and D inspections. The group B, C and D inspections stated in MIL-M-24791 shall apply.

4.6 Production representative device (PRD) screens and inspections. When specified in the acquisition documents (see 6.2), the inspection described in the subparagraphs to this paragraph shall be performed.

4.6.1 Screens. All device screens, as specified in MIL-M-24791, except for stabilization bake. The monitor procedure of the jitter measurement method of MIL-M-24791, Appendix B, shall be performed before and after temperature cycling, constant acceleration and burn-in screens, and during the burn-in screen. The filter bandwidth shall be 5 MHz. The change in the rms value of jitter between before and after measurements shall be no greater than that produced by a 0.5 dB change in input power before the test.

4.6.2 Group A inspection. The Group A inspection tests (see 4.5.1) at minimum, nominal, and maximum power supply voltages and at minimum, at 25 °C, and at maximum case operating temperature shall apply.

4.6.3 Group B inspection. The Group B inspection tests of MIL-M-24791 shall apply.

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4.6.4 Design performance inspection. All tests in Table V.

TABLE V. PRD design performance inspection.

Parameter	Requirement paragraph	Test paragraph
Electrical Interface	3.2.2	4.7.7.1
Transfer - Phasing	3.2.4.1.1	4.7.7.2
Optoelectronic Propagation Delay - Magnitude	3.2.4.1.3.1	4.7.7.3
Jitter - rms Random and Systematic	3.2.5.2.1	4.7.7.7
Jitter - Random and Systematic Spectral Density	3.2.5.2.2	4.7.7.8
Jitter - Pattern Dependent	3.2.5.3	4.7.7.9
Jitter Transfer - Envelope waveform	3.2.5.4.1	4.7.7.10
Jitter Transfer - From Power Supply Inputs	3.2.5.4.2	4.7.7.11

4.7 Test conditions and methods. Test methods, and conditions applicable to them and those in MIL-M-24791 are specified in the subparagraphs to this paragraph.

4.7.1 Operational test conditions. Certain test methods contain the reference "Operational test conditions apply." If so, then the operating conditions and acceptance criteria of the subparagraphs to this paragraph apply.

4.7.1.1 Device mounting. Certain test methods require that the method used to mount the device under test be identified. The device shall be mounted on a printed circuit board using either a socket or by soldering. Heat transfer shall be by natural convection; no auxiliary heat sinking shall be used. Sufficient clear space shall be provided around and above the device such that convection is not impaired and heat transfer between adjacent devices is insignificant.

4.7.1.2 Test circuit. The device shall be operated during testing as described in the jitter measurements test method of MIL-M-24791, Appendix B. Unless otherwise specified, the minimum power supply voltage (see 3.2.6.1) shall be used.

4.7.1.3 Acceptance criteria. The monitor procedure of the jitter measurement test method of MIL-M-24791, Appendix B, shall be performed. The filter bandwidth shall be 5 MHz. The rms value of jitter shall not exceed the lesser of that resulting from a reduction of 0.5 dB in average input power at the start of a test or the equivalent of the absolute magnitude specified by 3.2.5.2.1.

4.7.2 Package integrity test methods. All package integrity tests of MIL-M-24791, where indicated from the quality conformance inspections therein, apply without modification.

4.7.3 Stress tests. All stress tests of MIL-M-24791, where indicated from the quality conformance inspections therein, apply. Supplemental information required by certain of those tests is specified in the subparagraphs to this paragraph.

4.7.3.1 Burn-in. Operational test conditions apply. The "test condition" paragraph and its subparagraphs of MIL-STD-883, method 1015 do not apply. The test shall be performed at 85 °C +0/-2 °C for 240 hours. Post burn-in measurements shall be performed within 24 hours of completion of the test.

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4.7.3.2 Temperature cycling. Operational test condition apply except that devices shall not be operated during the test. The following test temperature extremes shall be substituted for those given in table I of MIL-STD-883, method 1010. The hot temperature shall be +85 ± 5 °C. The cold temperature shall be 0 ± 5 °C.

4.7.3.3 Steady state life. Operational test conditions apply. The "test duration", "test temperature" and "test conditions" paragraphs and their subparagraphs of MIL-STD-883, method 1005 do not apply. The test shall be performed at a minimum case temperature of 85°C for 2000 hours.

4.7.4 Optical connection test method. Operational test conditions apply, except for the acceptance criteria. The pigtail cable/case junction strength test method of MIL-M-24791 shall be performed and its acceptance criteria applies. The parameter to be monitored is jitter and shall be measured using the monitor test procedure described in the jitter measurements test method of MIL-M-24791, Appendix B.

4.7.5 Chemical test methods. All quality conformance inspection chemical tests of MIL-M-24791 apply. The subparagraph to this paragraph provides supplemental information required by the moisture resistance test method.

4.7.5.1 Moisture resistance. Operational test conditions apply. The minimum case temperature during step 7 of the test shall be 0 ± 2 °C.

4.7.6 Reliability test. The reliability test of MIL-M-24791 does not apply.

4.7.7 Functional. The methods to be employed for evaluating functional performance are specified in the subparagraphs to this paragraph.

4.7.7.1 Electrical interface.

4.7.7.1.1 DATA outputs. Each test method listed in table VI shall be performed separately on the DATA+ and DATA- outputs. Operational test conditions apply (see 4.7.1) except for acceptance criteria. The input envelope waveform shall be square wave which conforms to the output waveform parameters of MIL-M-24791/1 with power level of -31 dBm for type LS receiver and -35 dBm for type HS receivers. The load on each output shall be 50 Ω into V_{CC} - 2.0 Vdc. Acceptance criteria is stated in the condition column.

TABLE VI. Electrical interface test methods.

Parameter	Test method	Conditions
Transition times	MIL-STD-883, Method 3004	$t_{TLH} = t_{THL}$ = 2 ns maximum; Transition time measurement points: 20% and 80%; Frequency of input envelope waveform: 40 MBd
High Level Output Voltage	MIL-STD-883, Method 3006	Frequency of input envelope waveform: 125 MBd Step 3 of the method does not apply. V_{OH}min shall be measured (using an oscilloscope) and shall not be less than V_{CC} - 1.0 V within ± 15% of the center of pulse.
Low Level Output Voltage	MIL-STD-883, Method 3007	Frequency of input envelope waveform: 125 MBd Step 3 of the method does not apply. V_{OL}max shall be measured (using an oscilloscope) and shall not be greater than V_{CC} - 1.65 V within ± 15% of the center of pulse.

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4.7.7.1.2 Signal detect output. The digital receiver signal detect flag output procedure of the flag parameter measurements test method of MIL-M-24791, Appendix B, shall be performed. The test method for High Level Output Voltage in Table VI shall be performed with the signal detect in the "assert state" and Low Level Output Voltage test method of Table VI shall be performed with the signal detect in the "deassert" state during this test.

4.7.7.2 Transfer - phasing. The reconfiguration and phasing procedure for receivers of the optoelectronic delay measurement test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.3 Data optoelectronic propagation delay magnitude. The receiver delay magnitude procedure measurement of the optoelectronic delay test method of MIL-M-24791, Appendix B shall be performed.

4.7.7.4 Data optoelectronic propagation delay difference. The data optoelectronic propagation delay difference shall be determined from the bias distortion procedure for receivers of the modulation envelope waveform measurement test method of MIL-M-24791, Appendix B. The measured bias distortion is the data optoelectronic propagation delay.

4.7.7.5 Evaluation. The evaluation procedure of the correlation between BER and jitter test method of MIL-M-24791, Appendix B, shall be performed on all devices submitted for acceptance (see 4.5.1.1) of MIL-M-24791. The maximum of all values of dJ determined by 4.7.7.5.1 shall be used. The procedure shall be performed for both the sensitivity level of 3.2.4.1.2.3 and the improvement level of 3.2.4.1.2.4. The pattern sensitivity value determined in accordance with 4.7.7.5.2 shall apply.

4.7.7.5.1 Establishing correlation between jitter and BER. The direct measurement procedure of the correlation between BER and jitter test method of MIL-M-24791, Appendix B, shall be performed. The inferred procedure shall not be used. The number of devices tested shall be as designated by the procuring activity (see 4.5.1.1) of MIL-M-24791. The procedure shall be performed for both the sensitivity level of 3.2.4.1.2.3 and the improvement level of 3.2.4.1.2.4. The pattern sensitivity value determined in accordance with 4.7.5.5.2 shall apply.

4.7.7.5.2 Pattern sensitivity. Pattern sensitivity is to be determined using the pattern sensitivity procedure of the digital receiver BER measurement test method of MIL-M-24791, Appendix B. The number of devices tested shall be as designated by the procuring activity. If the maximum and minimum values of the range measured differ by not greater than 1 dB, the maximum value shall be used for subsequent tests. If the difference exceeds 1 dB then the maximum value may be discarded and the maximum of the remaining set of values used.

4.7.7.6 Signal detect. All digital receiver signal detect flag procedures of the flag response measurements method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.7 Rms random and systematic jitter. The random and systematic jitter measurements rms procedure for receivers from the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed. The jitter measurement bandwidth shall be as specified in 3.2.5.1.3.

4.7.7.8 Random and systematic jitter spectral density. The random and systematic jitter measurements spectral density procedure for receivers from the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed. The jitter measurement bandwidth shall be as specified in 3.2.5.2.2.

4.7.7.9 Pattern dependent jitter. The pattern dependent jitter magnitude measurement procedure for receivers of the jitter measurements test method of MIL-M-24791, Appendix B shall be performed.

4.7.7.10 Jitter transfer coefficient from envelope waveform to DATA output(s). The transfer measurements procedure for input to output for receivers of the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed.

4.7.7.11 Jitter transfer coefficient from power supply input(s) to DATA output(s). The transfer measurements procedure for power supply to output for receivers of the jitter measurements test method of MIL-M-24791, Appendix B, shall be performed.

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4.7.7.12 Power supply current. The power supply voltage and current test method for receivers of MIL-M-24791, Appendix B, shall be performed. The input signal shall be a square wave at 125 MBd. Operational conditions apply (see 4.7.1). The power supply current shall be measured after the determination that the monitor value of the jitter is within acceptable limits (see jitter measurements test method of MIL-M-24791, Appendix B).

5. PACKAGING

5.1 General. The packaging provisions of MIL-M-24791 shall apply.

6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. Digital fiber optic receivers (see 1.2) described herein are intended primarily for use in SAFENET local area network applications. See MIL-STD-2204 for information on SAFENET. MIL-HDBK-818-1 provides SAFENET network development and implementation guidance. They are also available for other applications which require digital fiber optic receivers with nominal data rates not exceeding 160 MBd.

6.2 Acquisition requirements. The acquisition requirements of MIL-M-24791 apply.

6.3 Definitions. Terms used in this specification are defined in the subparagraphs to this paragraph and in MIL-M-24791 and the documents cited therein for this purpose. Order of precedence is this paragraph followed by the order given in MIL-M-24791.

Note: If a definition is not shown here, see paragraph 6.3 of MIL-M-24791.

6.3.1 Jitter transfer coefficient. The ratio between a specified component of output jitter and the stimulus that produces it, either jitter of the input signal or other specified parameter.

6.3.2 Pattern dependent jitter. That component of jitter that is correlated with the pattern of the stimulus signal.

6.3.3 Random jitter. That component of jitter that is uncorrelated with any component of the stimulus signal or device transfer parameter.

6.3.4 Systematic jitter. That component of jitter that is correlated with a device transfer parameter.

6.4 Abbreviations. Abbreviations for terms or phrases used in this specification are listed below or found in MIL-M-24791 or the documents cited therein for this purpose. Order of precedence is this paragraph followed by the order given in MIL-M-24791.

a. BER: Bit error ratio or bit error rate

b. DODISS: Department of Defense Index of Specifications and Standards

c. LAN: Local area network

d. MTBF: Mean time between failures

e. QCI: Quality conformance inspection

f. SAFENET: Survivable Adaptable Fiber Optic Embedded Network

g. UI: Unit Interval

6.5 Symbols. Symbols used in this specification are listed below or found in MIL-M-24791 or the documents cited therein for this purpose. Order of precedence is this paragraph followed by the order given in MIL-M-24791.

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6.6 PIN. The Part or Identifying Number (PIN) shall be in accordance with MIL-M-24791. Type LS receiver shall have an -LS suffix added to the end of the part number; type HS shall have an -HS added to the end of the part number.

6.7 Approved source(s) of supply. The below approved source(s) of supply has submitted a quality assurance plan which has been reviewed and approved by the preparing activity, Naval Sea Systems Command.

Vendor cage number	Vendor name and address
0P926	Laser Diode, Incorporated Lightwave Products Division 4 Olsen Avenue Edison, NJ 08820

CONCLUDING MATERIALS

Custodian:
Navy - SH
Air Force - 85
Army - CR
NASA - NA
Review activities:
Navy - EC, OS, YD

DLA - ES

Preparing activity:
Navy - SH

Agent:
DLA - ES

(Project number: 6026-N003)

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height and the manufacturer's designating symbol shall be located

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-24791/2	2. DOCUMENT DATE (YYMMDD) 94/XX/XX

3. DOCUMENT TITLE

MODULE, FIBER OPTIC, RECEIVER, DIGITAL, 160 MBD, ASSOCIATED DETAIL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Kendall A. Cottongim	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-5373	(2) DSN 986-5373
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Naval Sea Systems Command SEA 05Q42, Washington, DC 20362-5160 Telephone (703) 602-9137/6020 DSN 332-9137/6020	

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the power ... f. The total device power dissipation at an ambient temperature of 25 ... above 25